

# Clinical Photo Capture Guide

## NEUROIJECT · SPASTICITY INJECTION GUIDE

30 shared position + 75 probe + 68 ultrasound = 173 images

 Patient Position    Probe + Needle Site    Ultrasound Image

**Patient position is shared** — many muscles set up identically, so the 75 muscles need only **30** position photos. Capture each **pos-<group>.jpg** once (table below) and every muscle in that group reuses it. Probe & ultrasound are per muscle (**<muscleId>-probe.jpg · -us.jpg**) in **assets/images/clinical/**. The probe + needle site is **one annotated photo**: a **blue bar** over the probe footprint and a **red dot** at the needle insertion point. **68/75** muscles are ultrasound-guided; for the rest the US image is optional and the site photo needs only the red dot.

### Patient positions · 30 photos, shot once & reused

|                                     |                                                                      |                                                                                                                                                                         |
|-------------------------------------|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| pos-supine-arm-abducted.jpg         | Supine · arm abducted, externally rotated                            | 2 · Pectoralis Major, Subscapularis                                                                                                                                     |
| pos-lateral-decub-arm-abducted.jpg  | Lateral decubitus (or prone) · arm slightly abducted                 | 1 · Latissimus Dorsi                                                                                                                                                    |
| pos-seated-or-prone-arm-at-side.jpg | Seated or prone · arm adducted at the side                           | 2 · Infraspinatus, Teres Major                                                                                                                                          |
| pos-arm-supinated-on-support.jpg    | Supine or seated · arm supinated on a support                        | 2 · Biceps Brachii, Brachialis                                                                                                                                          |
| pos-seated-elbow-flexed.jpg         | Seated or prone · elbow flexed ~90°                                  | 1 · Triceps                                                                                                                                                             |
| pos-seated-forearm-neutral.jpg      | Seated · forearm neutral, thumb up                                   | 1 · Brachioradialis                                                                                                                                                     |
| pos-seated-forearm-supinated.jpg    | Seated · forearm supinated on a flat surface                         | 7 · Pronator Teres, Flexor Carpi Radialis, Flexor Carpi Ulnaris, Flexor Digitorum Superficialis, Flexor Digitorum Profundus, Flexor Pollicis Longus, Pronator Quadratus |
| pos-seated-forearm-pronated.jpg     | Seated · forearm pronated on a flat surface                          | 4 · Extensor Carpi Radialis Longus, Extensor Carpi Radialis Brevis, Extensor Carpi Ulnaris, Extensor Digitorum Communis                                                 |
| pos-hand-pronated-flat.jpg          | Hand resting palm-down, thumb abducted                               | 1 · Adductor Pollicis                                                                                                                                                   |
| pos-hand-supinated.jpg              | Hand supinated, palm up, fingers relaxed                             | 1 · Hand Lumbricals                                                                                                                                                     |
| pos-prone-pillow-abdomen.jpg        | Prone · pillow under the abdomen                                     | 2 · Paraspinals (Erector Spinae Group), Quadratus Lumborum                                                                                                              |
| pos-supine-knees-bent.jpg           | Supine · knees slightly bent                                         | 1 · Rectus Abdominis                                                                                                                                                    |
| pos-supine-or-lat-decub.jpg         | Supine or lateral decubitus · affected side up                       | 3 · Obliques (External & Internal), Tensor Fasciae Latae (TFL), Vastus Lateralis                                                                                        |
| pos-supine-hip-neutral.jpg          | Supine · hip neutral, knee extended                                  | 1 · Iliopsoas                                                                                                                                                           |
| pos-prone-or-lat-decub-buttock.jpg  | Prone (or lateral decubitus) · buttock exposed                       | 1 · Gluteus Maximus                                                                                                                                                     |
| pos-prone-pillow-pelvis.jpg         | Prone · pillow under the pelvis                                      | 1 · Piriformis                                                                                                                                                          |
| pos-supine-frog-leg.jpg             | Supine · frog-leg (hip abducted & externally rotated)                | 4 · Pectineus, Adductor Longus, Adductor Brevis, Gracilis                                                                                                               |
| pos-prone-legs-apart.jpg            | Prone, legs apart (or supine, hip abducted & ER)                     | 1 · Adductor Magnus                                                                                                                                                     |
| pos-supine-knee-extended.jpg        | Supine · knee extended, thigh relaxed                                | 3 · Rectus Femoris, Vastus Medialis, Vastus Intermedius                                                                                                                 |
| pos-prone-knee-flexed.jpg           | Prone · knee slightly flexed (pillow/bolster)                        | 3 · Biceps Femoris, Semitendinosus, Semimembranosus                                                                                                                     |
| pos-prone-feet-off-bed.jpg          | Prone · feet hanging off the edge of the bed                         | 5 · Gastrocnemius (Medial), Gastrocnemius (Lateral), Soleus (Medial), Soleus (Lateral), Flexor Hallucis Longus (FHL)                                                    |
| pos-leg-ext-rotated.jpg             | Prone or supine · leg externally rotated (expose medial calf)        | 2 · Tibialis Posterior, Flexor Digitorum Longus (FDL)                                                                                                                   |
| pos-supine-leg-neutral.jpg          | Supine · leg extended, ankle neutral                                 | 3 · Tibialis Anterior, Extensor Hallucis Longus (EHL), Extensor Digitorum Longus                                                                                        |
| pos-supine-leg-internal-rot.jpg     | Supine · leg internally rotated (or lateral decubitus)               | 1 · Peroneus Longus                                                                                                                                                     |
| pos-expose-sole.jpg                 | Prone foot off bed (or supine, ankle dorsiflexed to expose the sole) | 3 · Flexor Digitorum Brevis (FDB), Adductor Hallucis, Foot Lumbricals                                                                                                   |
| pos-supine-head-neutral.jpg         | Supine · head neutral or turned slightly                             | 2 · Sternocleidomastoid (SCM), Scalenes (Anterior, Middle, Posterior)                                                                                                   |
| pos-seated-or-prone-forehead.jpg    | Seated upright, arms relaxed (or prone, forehead on hands)           | 2 · Upper Trapezius, Levator Scapulae                                                                                                                                   |
| pos-cervical-posterior-flexed.jpg   | Seated · neck slightly flexed (or prone, forehead on hands)          | 5 · Splenius Capitis, Splenius Cervicis, Obliquus Capitis Inferior, Semispinalis Capitis, Longissimus Capitis                                                           |
| pos-seated-face-forward.jpg         | Seated upright · face forward, relaxed                               | 6 · Frontalis, Corrugator Supercilii, Procerus, Orbicularis Oculi, Mentalis, Platysma                                                                                   |
| pos-seated-jaw-relaxed.jpg          | Seated upright (or supine) · jaw relaxed                             | 4 · Temporalis, Masseter, Lateral Pterygoid, Medial Pterygoid                                                                                                           |

## Pectoralis Major

 pec-major

Shoulder adduction, internal rotation

-  **Patient Position** pos-supine-arm-abducted.jpg  Position the patient supine with the arm abducted 30-45 degrees and externally rotated slightly to open the chest.
-  **Probe + Needle Site** pec-major-probe.jpg Patient supine, arm abducted 30°. Place probe transversely on anterior chest, lateral to sternum at 3rd-4th rib. Photo from... Needle: For the clavicular head: insert the needle 2-3 cm inferior to the lateral clavicle, angled slightly caudally at about 30 degrees...
-  **Ultrasound Image** pec-major-us.jpg Position the patient supine with the arm slightly abducted. Place the probe transversely on the anterior chest. Depth ≈ 5.5 cm.

## Latissimus Dorsi

 lat-dorsi

Shoulder adduction, internal rotation, extension

-  **Patient Position** pos-lateral-decub-arm-abducted.jpg  Position the patient in lateral decubitus with the affected side up, or prone with arm slightly abducted.
-  **Probe + Needle Site** lat-dorsi-probe.jpg Patient lateral decubitus or prone. Place probe transversely across posterior axillary fold. Photo from behind showing probe on... Needle: Insert the needle perpendicular to the skin into the mid-belly of the muscle.
-  **Ultrasound Image** lat-dorsi-us.jpg Identify subcutaneous tissue superficially as a hyperechoic layer.

## Subscapularis

 subscapularis

Shoulder internal rotation, Adduction

-  **Patient Position** pos-supine-arm-abducted.jpg  Position the patient supine or seated with the arm abducted to 45-60 degrees and externally rotated, resting on a support.
-  **Probe + Needle Site** subscapularis-probe.jpg Patient supine, arm abducted 45-60° and externally rotated. Place probe parasagittally in anterior axilla. Photo from front... Needle: Insert the needle through the ANTERIOR axillary fold (formed by pec major), directing it posterolaterally toward the anterior...
-  **Ultrasound Image** subscapularis-us.jpg Identify the subcutaneous fat and the pectoralis major as the most superficial muscular layer on the anterior chest.

## Infraspinatus

 infraspinatus

⚠️ NOT TYPICALLY INJECTED — External rotator (reference only)

-  **Patient Position** pos-seated-or-prone-arm-at-side.jpg  Position the patient seated, leaning forward, or in a prone position with the arms at the sides.
-  **Probe + Needle Site** infraspinatus-probe.jpg Patient seated leaning forward or prone. Place probe transversely across infraspinous fossa, parallel to scapular spine. Photo... Needle: Insert the needle perpendicular to the skin into the center of the infraspinous fossa, targeting the thickest part of the muscle...
-  **Ultrasound Image** infraspinatus-us.jpg Identify the subcutaneous fat and the thin trapezius muscle as the most superficial layers (the trapezius overlies the...)

## Teres Major

 teres-major

Shoulder adduction, Internal rotation, Extension

-  **Patient Position** pos-seated-or-prone-arm-at-side.jpg  Position the patient seated or prone with the arm adducted at the side.
-  **Probe + Needle Site** teres-major-probe.jpg Patient seated or prone, arm at side. Place probe transversely over posterior axillary fold. Photo from behind showing probe on... Needle: Identify the posterior axillary fold and the inferior angle of the scapula.
-  **Ultrasound Image** teres-major-us.jpg Position the patient seated or prone with the arm slightly abducted. Place the probe on the posterior axillary fold. Depth ≈ 3.5-4.5 cm.

## Biceps Brachii

 biceps-brachii

Elbow flexion, Forearm supination

-  **Patient Position** pos-arm-supinated-on-support.jpg  Position the patient supine or seated with the arm supinated and elbow extended, resting on a support surface.
-  **Probe + Needle Site** biceps-brachii-probe.jpg Patient supine, arm supinated. Place probe transversely on anterior mid-upper arm. Photo from lateral view showing probe across... Needle: Insert the needle into the bulk of the muscle belly at the mid-humeral level, perpendicular to the skin surface.
-  **Ultrasound Image** biceps-brachii-us.jpg Position the patient supine with the arm slightly abducted and externally rotated. Place the probe transversely on the anterior... Depth ≈ 3.5-4.0 cm.

## Brachialis

 brachialis

Pure elbow flexion

-  **Patient Position** pos-arm-supinated-on-support.jpg  Position the patient supine or seated with the elbow slightly flexed (about 30 degrees) and the forearm supinated.
-  **Probe + Needle Site** brachialis-probe.jpg Patient supine, elbow slightly flexed. Place probe transversely on anterior distal arm, 4-6 cm above elbow crease. Photo showing... Needle: Insert the needle perpendicular to the skin through the biceps belly, advancing into the brachialis below it.
-  **Ultrasound Image** brachialis-us.jpg Place the probe transversely across the anterior distal third of the arm, ~4-6 cm proximal to the elbow crease.

## Triceps

 triceps

Elbow extension

-  **Patient Position** pos-seated-elbow-flexed.jpg  Position the patient seated or prone with the arm resting on a support, elbow flexed to approximately 90 degrees.
-  **Probe + Needle Site** triceps-probe.jpg Patient seated or prone. Place probe transversely on posterior mid-upper arm. Photo from behind showing probe across triceps... Needle: For the long head: insert the needle into the medial portion of the posterior arm at mid-humeral level, perpendicular to the skin.
-  **Ultrasound Image** triceps-us.jpg Position the patient prone or seated with the arm supported. Place the probe transversely on the posterior arm. Depth ≈ 3.5-4.5 cm.

## Brachioradialis

 brachioradialis

Elbow flexion (forearm in neutral position)

-  **Patient Position** pos-seated-forearm-neutral.jpg  Position the patient seated with the forearm resting on a table in neutral position (thumb pointing upward).
-  **Probe + Needle Site** brachioradialis-probe.jpg Patient seated, forearm neutral on table. Place probe transversely on lateral proximal forearm, 4-6 cm below lateral epicondyle... Needle: Insert the needle into the bulk of the muscle belly at the junction of the proximal and middle thirds of the forearm,...
-  **Ultrasound Image** brachioradialis-us.jpg Place the probe transversely on the lateral forearm with the patient's elbow slightly flexed and forearm in neutral (thumb-up)... Depth ≈ 3.5 cm.

## Pronator Teres pronator-teres

Forearm pronation

-  **Patient Position** [pos-seated-forearm-supinated.jpg](#) shared  
Position the patient seated with the forearm supinated and the elbow extended, resting on a support surface.
-  **Probe + Needle Site** [pronator-teres-probe.jpg](#)  
Patient seated, forearm supinated. Place probe transversely on medial proximal forearm, 3-5 cm below medial epicondyle. Photo... Needle: Insert the needle 2-3 cm distal to the medial epicondyle, into the bulk of the muscle belly on the medial-to-anterior proximal...
-  **Ultrasound Image** [pronator-teres-us.jpg](#)  
Position the patient with the elbow extended and forearm supinated to stretch the pronator teres. Depth  $\approx$  4.5 cm.

## Flexor Carpi Radialis fcr

Wrist flexion, Radial deviation

-  **Patient Position** [pos-seated-forearm-supinated.jpg](#) shared  
Position the patient seated with the forearm supinated and resting on a flat surface.
-  **Probe + Needle Site** [fcr-probe.jpg](#)  
Patient seated, forearm supinated on table. Place probe transversely on central volar forearm at proximal-to-mid third. Photo... Needle: Insert the needle into the bulk of the muscle belly at the junction of the proximal and middle thirds of the forearm,...
-  **Ultrasound Image** [fcr-us.jpg](#)  
Place the probe transversely on the volar forearm, centered over the palpable FCR tendon. Depth  $\approx$  2.0 cm.

## Flexor Carpi Ulnaris fcu

Wrist flexion, Ulnar deviation

-  **Patient Position** [pos-seated-forearm-supinated.jpg](#) shared  
Position the patient seated with the forearm supinated or in neutral, resting on a support.
-  **Probe + Needle Site** [fcu-probe.jpg](#)  
Patient seated, forearm supinated. Place probe transversely on medial forearm. Photo showing probe on ulnar/medial side of... Needle: Insert the needle into the bulk of the muscle belly at the junction of the proximal and middle thirds of the forearm, on the...
-  **Ultrasound Image** [fcu-us.jpg](#)  
Place the probe transversely on the medial (ulnar) side of the forearm with the forearm fully supinated. Depth  $\approx$  2.5-3.5 cm.

## Flexor Digitorum Superficialis fds

Finger flexion at PIP joints, Wrist flexion

-  **Patient Position** [pos-seated-forearm-supinated.jpg](#) shared  
Position the patient seated with the forearm supinated and resting on a flat surface.
-  **Probe + Needle Site** [fds-probe.jpg](#)  
Patient seated, forearm supinated on table. Place probe transversely on volar forearm, 5-8 cm below medial epicondyle. Photo... Needle: Insert the needle at the mid-forearm level, between the FCR and FCU, slightly deeper than the superficial flexor layer.
-  **Ultrasound Image** [fds-us.jpg](#)  
Place the probe transversely on the central volar forearm. Depth  $\approx$  2.0-3.0 cm.

## Flexor Digitorum Profundus fdp

Finger flexion at DIP joints, Wrist flexion

-  **Patient Position** [pos-seated-forearm-supinated.jpg](#) shared  
Position the patient seated with the forearm supinated and resting on a flat surface.
-  **Probe + Needle Site** [fdp-probe.jpg](#)  
Patient seated, forearm supinated. Place probe transversely on medial-volar mid-forearm. Photo showing probe on deep volar... Needle: Insert the needle from the medial (ulnar) side of the forearm, just anterior to the palpable subcutaneous ulnar border, at the...
-  **Ultrasound Image** [fdp-us.jpg](#)  
Place the probe transversely on the volar-ulnar forearm at the mid-forearm level. Depth  $\approx$  3.0-4.0 cm.

## Flexor Pollicis Longus fpl

Thumb flexion at IP joint

-  **Patient Position** [pos-seated-forearm-supinated.jpg](#) shared  
Position the patient seated with the forearm supinated and resting flat on a support surface.
-  **Probe + Needle Site** [fpl-probe.jpg](#)  
Patient seated, forearm supinated flat. Place probe transversely on mid-volar forearm at junction of middle and distal thirds.... Needle: Insert the needle at the mid-forearm level, on the radial side, between the brachioradialis (laterally) and the FCR (medially).
-  **Ultrasound Image** [fpl-us.jpg](#)  
Place the probe transversely on the volar forearm with the patient supinated. Depth  $\approx$  3.0-3.5 cm.

## Pronator Quadratus pronator-quadratus

Forearm pronation (distal)

-  **Patient Position** [pos-seated-forearm-supinated.jpg](#) shared  
Position the patient supine or seated with the forearm supinated and resting on a support, wrist in neutral.
-  **Probe + Needle Site** [pronator-quadratus-probe.jpg](#)  
Patient supine, forearm supinated on table. Place probe transversely on volar distal forearm, 3-5 cm proximal to wrist crease.... Needle: DORSAL APPROACH (preferred): Insert the needle from the dorsal distal forearm, between the radius and ulna, advancing through the...
-  **Ultrasound Image** [pronator-quadratus-us.jpg](#)  
Place the probe transversely on the volar distal forearm, 3-5 cm proximal to the wrist crease.

## Extensor Carpi Radialis Longus ecrl

Wrist extension, Radial deviation

-  **Patient Position** [pos-seated-forearm-pronated.jpg](#) shared  
Position the patient seated with the forearm pronated and resting on a flat surface to expose the dorsoradial forearm.
-  **Probe + Needle Site** [ecrl-probe.jpg](#)  
Patient seated, forearm pronated on table. Place probe transversely on the radiodorsal proximal forearm, 3-5 cm below the lateral... Needle: Insert the needle into the muscle belly in the proximal third of the forearm, perpendicular to the skin.
-  **Ultrasound Image** [ecrl-us.jpg](#)  
Place the probe transversely on the dorsoradial proximal forearm with the forearm pronated. Depth  $\approx$  1-2 cm.

## Extensor Carpi Radialis Brevis ecrb

Wrist extension

-  **Patient Position** [pos-seated-forearm-pronated.jpg](#) shared  
Position the patient seated with the forearm pronated and resting on a flat surface.
-  **Probe + Needle Site** [ecrb-probe.jpg](#)  
Patient seated, forearm pronated. Place probe transversely on the radiodorsal proximal forearm, 4-6 cm below the lateral... Needle: Insert the needle into the proximal third of the forearm, angling slightly deeper than the ECRL.
-  **Ultrasound Image** [ecrb-us.jpg](#)  
Place the probe transversely on the dorsoradial proximal forearm, forearm pronated. Depth  $\approx$  1.5-2.5 cm.

## Extensor Carpi Ulnaris ecu

Wrist extension, Ulnar deviation

-  **Patient Position** [pos-seated-forearm-pronated.jpg](#) shared  
Position the patient seated with the forearm pronated and resting on a flat surface to expose the dorsal forearm.
-  **Probe + Needle Site** [ecu-probe.jpg](#)  
Patient seated, forearm pronated. Place probe transversely on the dorsoulnar forearm, just radial to the subcutaneous ulnar... Needle: Insert the needle into the muscle belly in the proximal-to-mid forearm, perpendicular to the skin.
-  **Ultrasound Image** [ecu-us.jpg](#)  
Place the probe transversely on the dorsoulnar forearm with the forearm pronated. Depth  $\approx$  1-1.5 cm.

## Extensor Digitorum Communis edc

*Finger extension (MCP)*

-  **Patient Position** [pos-seated-forearm-pronated.jpg](#) shared  
Position the patient seated with the forearm pronated and resting on a flat surface.
-  **Probe + Needle Site** [edc-probe.jpg](#)  
Patient seated, forearm pronated. Place probe transversely on the central dorsal proximal-to-mid forearm, between the radial... Needle: Insert the needle into the muscle belly in the proximal-to-mid forearm, perpendicular to the skin.
-  **Ultrasound Image** [edc-us.jpg](#)  
Place the probe transversely on the central dorsal forearm with the forearm pronated. Depth  $\approx$  1-1.5 cm.

## Adductor Pollicis adductor-pollicis

*Thumb adduction (thumb-in-palm deformity)*

-  **Patient Position** [pos-hand-pronated-flat.jpg](#) shared  
Position the patient with the hand resting on a flat surface, palm down (pronated), with the thumb abducted away from the palm.
-  **Probe + Needle Site** [adductor-pollicis-probe.jpg](#)  
Patient hand pronated on table. Place small linear or hockey-stick probe transversely across first web space from dorsal... Needle: Insert the needle from the dorsal aspect of the first web space, perpendicular to the plane of the hand.
-  **Ultrasound Image** [adductor-pollicis-us.jpg](#)  
Identify the subcutaneous fat and thin dorsal skin as the most superficial layers.

## Hand Lumbricals hand-lumbricals

*MCP flexion + IP extension (intrinsic-plus posture)*

-  **Patient Position** [pos-hand-supinated.jpg](#) shared  
Position the patient with the hand supinated (palm up), fingers relaxed and slightly extended.
-  **Probe + Needle Site** [hand-lumbricals-probe.jpg](#)  
Patient palm-up, fingers relaxed. Place hockey-stick probe longitudinally on palm along the metacarpal. Photo from palmar view... Needle: PALMAR APPROACH: Insert the needle into the palm just proximal to the MCP joint of the target finger, on the radial side of the...
-  **Ultrasound Image** [hand-lumbricals-us.jpg](#)  
Place the probe longitudinally on the palmar surface along the metacarpal of the target finger.

## Lower Extremity · 34 muscles · 102 photos

### Paraspinals (Erector Spinae Group) paraspinals

*Trunk extension spasticity, axial dystonia, camptocormia treatment*

-  **Patient Position** [pos-prone-pillow-abdomen.jpg](#) shared  
Position the patient prone with a pillow under the abdomen to reduce lumbar lordosis and open the interlaminar spaces.
-  **Probe + Needle Site** [paraspinals-probe.jpg](#)  
Patient prone, pillow under abdomen. Place curvilinear probe parasagittally 2-3 cm lateral to midline spinous processes, or... Needle: Insert the needle perpendicular to the skin or at a slight lateral-to-medial angle, advancing into the muscle belly.
-  **Ultrasound Image** [paraspinals-us.jpg](#)  
PARASAGITTAL VIEW: Place the probe longitudinally (parasagittal), 2-3 cm lateral to the midline spinous processes. The transverse...

### Obliques (External & Internal) obliques

*Trunk rotation, lateral flexion*

-  **Patient Position** [pos-supine-or-lat-decub.jpg](#) shared  
Position the patient supine or in the lateral decubitus position with the target side up.
-  **Probe + Needle Site** [obliques-probe.jpg](#)  
Patient supine or lateral decubitus. Place linear probe transversely on the lateral abdominal wall between the costal margin and... Needle: Target the external oblique and/or internal oblique depending on the clinical pattern — typically inject the internal oblique as...
-  **Ultrasound Image** [obliques-us.jpg](#)  
Place the probe transversely on the lateral abdominal wall at the level of the umbilicus, along the mid-axillary line.

### Quadratus Lumborum quadratus-lumborum

*Pelvic hiking, trunk lateral flexion*

-  **Patient Position** [pos-prone-pillow-abdomen.jpg](#) shared  
Position the patient prone with a pillow under the abdomen to reduce lumbar lordosis.
-  **Probe + Needle Site** [quadratus-lumborum-probe.jpg](#)  
Patient prone, pillow under abdomen. Place curvilinear probe transversely on flank, just above iliac crest. Photo from side... Needle: Insert the needle perpendicular to the skin, angling slightly medially toward the transverse processes of L3/L4.
-  **Ultrasound Image** [quadratus-lumborum-us.jpg](#)  
Identify the subcutaneous fat and thoracolumbar fascia superficially as hyperechoic layers.

### Iliopsoas iliopsoas

*Hip flexion*

-  **Patient Position** [pos-supine-hip-neutral.jpg](#) shared  
Position the patient supine with the hip in neutral position and the knee extended.
-  **Probe + Needle Site** [iliopsoas-probe.jpg](#)  
Patient supine, hip neutral. Place curvilinear probe transversely on proximal anterior thigh, 2-3 cm below inguinal ligament,... Needle: Under ultrasound guidance, insert the needle lateral to the femoral vessels and advance toward the iliopsoas muscle belly.
-  **Ultrasound Image** [iliopsoas-us.jpg](#)  
Identify subcutaneous fat, the fascia lata, and the Sartorius muscle as the most superficial structures laterally.

### Rectus Abdominis rectus-abdominis

*Trunk flexion (camptocormia / forward-flexed posture)*

-  **Patient Position** [pos-supine-knees-bent.jpg](#) shared  
Position the patient supine with the knees slightly bent to relax the abdominal wall.
-  **Probe + Needle Site** [rectus-abdominis-probe.jpg](#)  
Patient supine, knees bent. Place linear probe transversely on the anterior abdomen, just lateral to midline at the umbilical... Needle: Inject at 2-4 sites distributed along the length of the muscle — typically at the supra-umbilical and infra-umbilical levels...
-  **Ultrasound Image** [rectus-abdominis-us.jpg](#)  
Place the probe transversely on the anterior abdominal wall, just lateral to the midline at the level of the umbilicus.

### Gluteus Maximus gluteus-maximus

*Hip extension, external rotation*

-  **Patient Position** [pos-prone-or-lat-decub-buttock.jpg](#) shared  
Position the patient prone or in lateral decubitus with the affected side up.
-  **Probe + Needle Site** [gluteus-maximus-probe.jpg](#)  
Patient prone. Place curvilinear probe transversely on upper-outer quadrant of buttock. Photo from behind showing probe position... Needle: Insert the needle perpendicular to the skin into the muscle bulk at multiple sites distributed across the belly (3-4 sites for a...
-  **Ultrasound Image** [gluteus-maximus-us.jpg](#)  
Identify the thick subcutaneous fat layer, which can be substantial in this region.

## Piriformis piriformis

*Hip external rotation, abduction*

 **Patient Position** [pos-prone-pillow-pelvis.jpg](#) shared  
Position the patient prone with a pillow under the pelvis to slightly flex the hips.

 **Probe + Needle Site** [piriformis-probe.jpg](#)  
Patient prone, pillow under pelvis. Place curvilinear probe transversely along PSIS-to-greater-trochanter line, at midpoint.... Needle: Insert the needle perpendicular to the skin surface at the midpoint.

 **Ultrasound Image** [piriformis-us.jpg](#)  
Identify the subcutaneous tissue and the thick Gluteus Maximus as a large superficial hypoechoic muscle.

## Tensor Fasciae Latae (TFL) tfl

*Hip flexion, abduction, internal rotation*

 **Patient Position** [pos-supine-or-lat-decub.jpg](#) shared  
Position the patient supine or in lateral decubitus with the affected side up.

 **Probe + Needle Site** [tfl-probe.jpg](#)  
Patient supine or lateral. Place probe transversely on anterolateral hip, 2-3 cm below ASIS. Photo showing probe just distal to.... Needle: Insert the needle perpendicular to the skin surface directly into the palpated muscle belly.

 **Ultrasound Image** [tfl-us.jpg](#)  
Identify the subcutaneous fat as a superficial hyperechoic layer — it is usually thin in this region.

## Pectineus pectineus

*Hip flexion, Hip adduction*

 **Patient Position** [pos-supine-frog-leg.jpg](#) shared  
Position the patient supine with the hip slightly abducted and externally rotated (frog-leg) to open the groin and femoral triangle.

 **Probe + Needle Site** [pectineus-probe.jpg](#)  
Patient supine, hip abducted/externally rotated. Place probe transversely in the groin just below the inguinal crease; identify.... Needle: Insert the needle from MEDIAL to lateral into the pectineus belly, keeping the femoral vessels in view at all times.

 **Ultrasound Image** [pectineus-us.jpg](#)  
Place the probe transversely in the groin just below the inguinal crease and identify the femoral artery and vein with color.... Depth  $\approx$  1.5-3 cm.

## Adductor Longus adductor-longus

*Hip adduction*

 **Patient Position** [pos-supine-frog-leg.jpg](#) shared  
Position the patient supine with the hip slightly abducted and externally rotated, and the knee flexed with the sole of the foot resting against the...

 **Probe + Needle Site** [adductor-longus-probe.jpg](#)  
Patient supine, hip slightly abducted and externally rotated. Place probe transversely on proximal medial thigh, 3-5 cm below.... Needle: Insert the needle perpendicular to the skin into the muscle belly just lateral to the palpable tendon.

 **Ultrasound Image** [adductor-longus-us.jpg](#)  
Identify subcutaneous fat and the fascia lata as the superficial layers.

## Adductor Brevis adductor-brevis

*Hip adduction (deep, short adductor)*

 **Patient Position** [pos-supine-frog-leg.jpg](#) shared  
Position the patient supine with the hip abducted 30-45 degrees, knee flexed, and the leg externally rotated (frog-leg position).

 **Probe + Needle Site** [adductor-brevis-probe.jpg](#)  
Patient supine, frog-leg position. Place probe transversely on proximal medial thigh at the adductor longus origin. Photo from.... Needle: USG is strongly recommended — insert the needle through the adductor longus to reach the adductor brevis layer deep to it.

 **Ultrasound Image** [adductor-brevis-us.jpg](#)  
Place the probe transversely on the proximal medial thigh with the hip abducted.

## Adductor Magnus adductor-magnus

*Hip adduction, Hip extension*

 **Patient Position** [pos-prone-legs-apart.jpg](#) shared  
Position the patient prone with the legs slightly apart, or supine with the hip abducted and externally rotated.

 **Probe + Needle Site** [adductor-magnus-probe.jpg](#)  
Patient prone, legs slightly apart. Place probe transversely on posteromedial thigh at junction of middle and distal thirds.... Needle: Insert the needle from the posteromedial aspect of the thigh, angling slightly anteriorly.

 **Ultrasound Image** [adductor-magnus-us.jpg](#)  
Identify subcutaneous fat and the deep fascia as the superficial layers.

## Gracilis gracilis

*Hip adduction + knee flexion (crosses both joints)*

 **Patient Position** [pos-supine-frog-leg.jpg](#) shared  
Position the patient supine with the hip abducted 30-45 degrees and knee slightly flexed.

 **Probe + Needle Site** [gracilis-probe.jpg](#)  
Patient supine, frog-leg position. Place probe transversely on proximal medial thigh. Photo from medial view showing probe on.... Needle: Insert the needle perpendicular to the skin on the medial thigh into the superficial muscle belly.

 **Ultrasound Image** [gracilis-us.jpg](#)  
Place the probe transversely on the proximal medial thigh with the hip abducted.

## Rectus Femoris rectus-femoris

*Stiff knee gait, Knee extension*

 **Patient Position** [pos-supine-knee-extended.jpg](#) shared  
Position the patient supine with the knee extended and the thigh relaxed.

 **Probe + Needle Site** [rectus-femoris-probe.jpg](#)  
Patient supine, knee extended. Place probe transversely on anterior mid-thigh. Photo from lateral view showing probe across.... Needle: Insert the needle perpendicular to the skin into the bulk of the muscle belly at mid-thigh.

 **Ultrasound Image** [rectus-femoris-us.jpg](#)  
Identify subcutaneous fat and the superficial fascia (fascia lata) as the most superficial hyperechoic layers.

## Vastus Lateralis vastus-lateralis

*Knee extension*

 **Patient Position** [pos-supine-or-lat-decub.jpg](#) shared  
Position the patient supine or in lateral decubitus with the affected side up.

 **Probe + Needle Site** [vastus-lateralis-probe.jpg](#)  
Patient supine or lateral. Place probe transversely on lateral mid-thigh. Photo from lateral view showing probe on outer thigh. Needle: Insert the needle through the IT band, angling slightly anteriorly to enter the VL muscle belly.

 **Ultrasound Image** [vastus-lateralis-us.jpg](#)  
Identify subcutaneous fat and the fascia lata (including the IT band component) as hyperechoic superficial layers.

## Vastus Medialis vastus-medialis

*Knee extension (distal stabilization)*

 **Patient Position** [pos-supine-knee-extended.jpg](#) shared  
Position the patient supine with the knee extended and the thigh relaxed.

 **Probe + Needle Site** [vastus-medialis-probe.jpg](#)  
Patient supine, knee extended. Place probe transversely on distal medial thigh, just above patella (VMO area). Photo showing.... Needle: Insert the needle perpendicular to the skin into the bulk of the VMO muscle belly.

 **Ultrasound Image** [vastus-medialis-us.jpg](#)  
Identify subcutaneous tissue and the superficial fascia as hyperechoic layers.

## Vastus Intermedius vastus-intermedius

Knee extension

 **Patient Position** pos-supine-knee-extended.jpg 

Position the patient supine with the knee extended and the thigh relaxed.

 **Probe + Needle Site** vastus-intermedius-probe.jpg

Patient supine, knee extended. Place probe transversely on anterior proximal thigh (same region as rectus femoris but deeper... Needle: Insert the needle perpendicular to the skin through the center of the Rectus Femoris.

 **Ultrasound Image** vastus-intermedius-us.jpg

Identify subcutaneous fat and fascia lata superficially.

## Biceps Femoris biceps-femoris

Knee flexion, Hip extension

 **Patient Position** pos-prone-knee-flexed.jpg 

Position the patient prone with the knee slightly flexed over a pillow or bolster.

 **Probe + Needle Site** biceps-femoris-probe.jpg

Patient prone, knee flexed over pillow. Place probe transversely on lateral posterior thigh at junction of proximal and middle... Needle: Insert the needle perpendicular to the skin into the lateral muscle belly of the posterior thigh.

 **Ultrasound Image** biceps-femoris-us.jpg

Identify subcutaneous fat and the deep fascia (fascia lata) as the superficial layers.

## Semitendinosus semitendinosus

Knee flexion

 **Patient Position** pos-prone-knee-flexed.jpg 

Position the patient prone with the knee slightly flexed (place a pillow under the ankle).

 **Probe + Needle Site** semitendinosus-probe.jpg

Patient prone, knee slightly flexed. Place probe transversely on posterior mid-thigh. Photo from behind showing probe on... Needle: Insert the needle perpendicular to the skin into the mid-belly of the muscle (proximal to the tendinous portion).

 **Ultrasound Image** semitendinosus-us.jpg

Identify subcutaneous tissue and the deep fascia superficially.

## Semimembranosus semimembranosus

Knee flexion

 **Patient Position** pos-prone-knee-flexed.jpg 

Position the patient prone with the knee slightly flexed (place a pillow under the ankle).

 **Probe + Needle Site** semimembranosus-probe.jpg

Patient prone, knee slightly flexed. Place probe transversely on posterior medial thigh at mid-to-distal level. Photo from behind... Needle: Insert the needle from the medial aspect, angling slightly posteriorly and laterally to pass deep to the Semitendinosus.

 **Ultrasound Image** semimembranosus-us.jpg

Identify subcutaneous tissue and the deep fascia as superficial hyperechoic layers.

## Gastrocnemius (Medial) gastrocnemius-medial

Equinus deformity, Plantarflexion

 **Patient Position** pos-prone-feet-off-bed.jpg 

Position the patient prone with the feet hanging off the edge of the bed.

 **Probe + Needle Site** gastrocnemius-medial-probe.jpg

Patient prone, feet off bed edge. Place probe transversely on proximal medial calf, 2-3 cm below popliteal crease. Photo from... Needle: Insert the needle perpendicular to the skin into the bulk of the medial gastrocnemius belly.

 **Ultrasound Image** gastrocnemius-medial-us.jpg

Identify subcutaneous fat and the superficial fascia.

## Gastrocnemius (Lateral) gastrocnemius-lateral

Equinus deformity, Plantarflexion

 **Patient Position** pos-prone-feet-off-bed.jpg 

Position the patient prone with the feet hanging off the edge of the bed.

 **Probe + Needle Site** gastrocnemius-lateral-probe.jpg

Patient prone, feet off bed edge. Place probe transversely on proximal lateral calf, 2-3 cm below popliteal crease. Photo from... Needle: Insert the needle perpendicular to the skin into the lateral muscle belly.

 **Ultrasound Image** gastrocnemius-lateral-us.jpg

Identify subcutaneous fat and superficial fascia.

## Soleus (Medial) soleus-medial

Plantarflexion

 **Patient Position** pos-prone-feet-off-bed.jpg 

Position the patient prone with the feet hanging off the edge of the bed.

 **Probe + Needle Site** soleus-medial-probe.jpg

Patient prone, feet off edge. Place probe transversely on medial mid-calf, just posterior to tibial border. Photo from... Needle: Insert the needle perpendicular to the skin, directed slightly laterally and posteriorly.

 **Ultrasound Image** soleus-medial-us.jpg

Identify subcutaneous tissue and superficial fascia.

## Soleus (Lateral) soleus-lateral

Plantarflexion

 **Patient Position** pos-prone-feet-off-bed.jpg 

Position the patient prone with the feet hanging off the edge of the bed.

 **Probe + Needle Site** soleus-lateral-probe.jpg

Patient prone, feet off edge. Place probe transversely on lateral mid-calf. Photo from behind showing probe on lateral mid-calf. Needle: Insert the needle perpendicular to the skin into this lateral soleus bulge at mid-calf level.

 **Ultrasound Image** soleus-lateral-us.jpg

Identify subcutaneous tissue and superficial fascia.

## Tibialis Posterior tibialis-posterior

Equinovarus (Plantarflexion + Inversion)

 **Patient Position** pos-leg-ext-rotated.jpg 

Position the patient prone or supine with the leg externally rotated.

 **Probe + Needle Site** tibialis-posterior-probe.jpg

Patient prone or supine with leg externally rotated. Place probe transversely on medial mid-leg, just behind tibia. Photo from... Needle: Insert the needle directed posterolaterally, sliding along the posterior aspect of the tibia.

 **Ultrasound Image** tibialis-posterior-us.jpg

Identify subcutaneous tissue and the crural fascia superficially.

## Flexor Digitorum Longus (FDL) fdl

Claw toes, toe flexion

 **Patient Position** pos-leg-ext-rotated.jpg 

Position the patient prone or supine with the leg externally rotated to expose the medial calf.

 **Probe + Needle Site** fdl-probe.jpg

Patient prone or supine with leg externally rotated. Place probe transversely on medial distal leg, just behind tibial border.... Needle: Insert the needle perpendicular to the skin, directing it toward the posterior aspect of the tibia and fibula.

 **Ultrasound Image** fdl-us.jpg

Identify subcutaneous tissue and the superficial crural fascia.



## Flexor Hallucis Longus (FHL) fhl

*Claw hallux, Equinovarus support*

☒ **Patient Position** [pos-prone-feet-off-bed.jpg](#) shared

Position the patient prone with the feet hanging off the edge of the bed.

☐ **Probe + Needle Site** [fhl-probe.jpg](#)

Patient prone, feet off edge. Place probe transversely on distal posterolateral calf, adjacent to fibula. Photo from behind... Needle: Insert the needle lateral to the Achilles tendon, directed toward the posterior fibula.

☐ **Ultrasound Image** [fhl-us.jpg](#)

Identify subcutaneous tissue and the Soleus muscle/aponeurosis superficially.

## Tibialis Anterior tibialis-anterior

⚠ *NOT TYPICALLY INJECTED — Dorsiflexor (reference only)*

☒ **Patient Position** [pos-supine-leg-neutral.jpg](#) shared

Position the patient supine with the leg extended and the ankle in neutral.

☐ **Probe + Needle Site** [tibialis-anterior-probe.jpg](#)

Patient supine, leg extended. Place probe transversely on proximal anterolateral leg, 2 cm lateral to tibial crest. Photo from... Needle: Insert the needle perpendicular to the skin into the largest muscle bulk, approximately 2 cm lateral to the tibial crest.

☐ **Ultrasound Image** [tibialis-anterior-us.jpg](#)

Identify subcutaneous tissue — it is thin on the anterior leg.

## Extensor Hallucis Longus (EHL) ehl

*Hallux extension, hitchhiking thumb sign*

☒ **Patient Position** [pos-supine-leg-neutral.jpg](#) shared

Position the patient supine with the leg extended and the ankle in neutral.

☐ **Probe + Needle Site** [ehl-probe.jpg](#)

Patient supine, leg extended. Place probe transversely on distal anterolateral leg, ~10 cm above ankle. Photo from front showing... Needle: Insert the needle perpendicular to the skin into the EHL muscle belly.

☐ **Ultrasound Image** [ehl-us.jpg](#)

Identify the thin subcutaneous tissue on the anterior leg.

## Extensor Digitorum Longus edl

*Toe extension, Ankle dorsiflexion*

☒ **Patient Position** [pos-supine-leg-neutral.jpg](#) shared

Position the patient supine with the leg in neutral position and the foot relaxed.

☐ **Probe + Needle Site** [edl-probe.jpg](#)

Patient supine, leg neutral. Place probe transversely on anterolateral proximal leg at proximal-middle third junction. Photo from... Needle: Insert the needle perpendicular to the skin into the anterolateral muscle belly.

☐ **Ultrasound Image** [edl-us.jpg](#)

Identify subcutaneous fat and the crural fascia as the superficial layers.

## Peroneus Longus peroneus-longus

*Ankle eversion, Ankle plantarflexion*

☒ **Patient Position** [pos-supine-leg-internal-rot.jpg](#) shared

Position the patient supine with the leg internally rotated to expose the lateral compartment, or in a lateral decubitus position with the affected...

☐ **Probe + Needle Site** [peroneus-longus-probe.jpg](#)

Patient supine, leg internally rotated. Place probe transversely on lateral proximal leg, 5-8 cm below fibular head. Photo from... Needle: Insert the needle perpendicular to the skin into the lateral muscle belly, aiming toward the fibula.

☐ **Ultrasound Image** [peroneus-longus-us.jpg](#)

Identify subcutaneous fat and the crural fascia as the superficial layers.

## Flexor Digitorum Brevis (FDB) fdb

*Toe flexion (plantar)*

☒ **Patient Position** [pos-expose-sole.jpg](#) shared

Position the patient prone with the foot hanging off the edge of the bed, or supine with the ankle dorsiflexed to expose the sole.

☐ **Probe + Needle Site** [fdb-probe.jpg](#)

Patient prone, foot hanging off bed. Place probe longitudinally on plantar midfoot from calcaneus toward toes. Photo from plantar... Needle: Insert the needle perpendicular to the plantar skin surface.

☐ **Ultrasound Image** [fdb-us.jpg](#)

Identify the thick hyperechoic plantar fascia as the most superficial fibrous layer on the sole.

## Adductor Hallucis add-hallucis

*Hallux adduction, transverse arch collapse*

☒ **Patient Position** [pos-expose-sole.jpg](#) shared

Position the patient supine with the foot accessible, or prone with the foot dorsiflexed.

☐ **Probe + Needle Site** [add-hallucis-probe.jpg](#)

Patient supine, foot accessible. Place probe longitudinally along 1st-2nd intermetatarsal space on plantar surface. Photo showing... Needle: A dorsal approach is an alternative: insert the needle from the dorsum of the foot between the 1st and 2nd metatarsal shafts,...

☐ **Ultrasound Image** [add-hallucis-us.jpg](#)

Identify the hyperechoic cortex of the 1st and 2nd metatarsal shafts as bright lines with posterior shadowing.

## Foot Lumbricals foot-lumbricals

*MTP flexion + IP extension in the foot (contributes to claw toe deformity)*

☒ **Patient Position** [pos-expose-sole.jpg](#) shared

Position the patient prone with the foot hanging off the bed, or supine with the ankle dorsiflexed to expose the sole.

☐ **Probe + Needle Site** [foot-lumbricals-probe.jpg](#)

Patient prone or supine with foot exposed. Place hockey-stick probe longitudinally on plantar surface along the metatarsal. Photo... Needle: Insert the needle from the plantar surface perpendicular to the skin, just proximal to the MTP joint of the target toe, on the...

☐ **Ultrasound Image** [foot-lumbricals-us.jpg](#)

Place the probe longitudinally on the plantar surface along the metatarsal of the target toe.

## Sternocleidomastoid (SCM) scm

Head rotation to opposite side, ipsilateral lateral flexion

-  **Patient Position** pos-supine-head-neutral.jpg  Position the patient supine with the head turned slightly to the contralateral side to make the SCM prominent but not maximally stretched.
-  **Probe + Needle Site** scm-probe.jpg Patient supine, head turned slightly contralateral. Place probe transversely on lateral neck at thyroid cartilage level. Photo... Needle: Palpate the muscle belly at the mid-cervical level (approximately the level of the thyroid cartilage). Grasp the muscle between...
-  **Ultrasound Image** scm-us.jpg Identify the skin and subcutaneous fat as the most superficial hyperechoic layer.

## Scalenes (Anterior, Middle, Posterior) scalenes

Ipsilateral lateral neck flexion (laterocollis), accessory inspiration

-  **Patient Position** pos-supine-head-neutral.jpg  Position the patient supine with the head in neutral position or turned slightly to the contralateral side to expose the lateral neck.
-  **Probe + Needle Site** scalenes-probe.jpg Patient supine, head neutral or slightly turned contralateral. Place probe transversely on lateral neck at C5-C7 level, posterior... Needle: Under ultrasound, identify the target scalene muscle and confirm the location of the brachial plexus, subclavian/vertebral...
-  **Ultrasound Image** scalenes-us.jpg Identify the SCM as the most superficial anterolateral muscle — use it as your anterior landmark.

## Upper Trapezius upper-trapezius

Scapular elevation, head lateral flexion

-  **Patient Position** pos-seated-or-prone-forehead.jpg  Position the patient seated upright with arms relaxed at the sides, or prone with the forehead resting on hands.
-  **Probe + Needle Site** upper-trapezius-probe.jpg Patient seated, arms relaxed. Place probe transversely on upper trapezius ridge, midway between cervical spine and acromion.... Needle: Insert the needle perpendicular to the skin surface into the midpoint of the palpated muscle bulk, advancing 1-2 cm into the...
-  **Ultrasound Image** upper-trapezius-us.jpg Identify the skin and subcutaneous fat as the most superficial hyperechoic layer.

## Levator Scapulae levator-scapulae

Scapular elevation, cervical lateral flexion and rotation

-  **Patient Position** pos-seated-or-prone-forehead.jpg  Position the patient seated upright with arms relaxed, or prone with the forehead resting on hands. Slight contralateral lateral flexion of the neck...
-  **Probe + Needle Site** levator-scapulae-probe.jpg Patient seated, arms relaxed. Place probe transversely over superior angle of scapula, 2-3 cm lateral to C7-T2 spinous processes.... Needle: Identify the superior angle of the scapula by palpating the superomedial corner of the scapula — this is the most reliable...
-  **Ultrasound Image** levator-scapulae-us.jpg Identify the skin and subcutaneous fat as the most superficial hyperechoic layer.

## Splenius Capitis splenius-capitis

Ipsilateral head rotation and extension

-  **Patient Position** pos-cervical-posterior-flexed.jpg  Position the patient seated upright with the neck slightly flexed to open the posterior cervical space, or prone with the forehead resting on hands.
-  **Probe + Needle Site** splenius-capitis-probe.jpg Patient seated, neck slightly flexed. Place probe transversely on posterolateral neck at C2-C4 level, 2-3 cm lateral to midline.... Needle: Insert the needle perpendicular to the skin, advancing through the trapezius (1-1.5 cm) and into the splenius capitis beneath it...
-  **Ultrasound Image** splenius-capitis-us.jpg Identify the skin and subcutaneous fat as the most superficial hyperechoic layer.

## Splenius Cervicis splenius-cervicis

Cervical dystonia — ipsilateral rotation & laterocollis

-  **Patient Position** pos-cervical-posterior-flexed.jpg  Position the patient prone or seated with the neck slightly flexed and the forehead supported.
-  **Probe + Needle Site** splenius-cervicis-probe.jpg Patient prone/seated, neck slightly flexed. Place probe transversely on the posterolateral neck at C2-C4; splenius cervicis lies... Needle: Insert the needle in-plane toward the muscle belly, staying lateral to the laminae and superficial to the transverse processes.
-  **Ultrasound Image** splenius-cervicis-us.jpg Place the probe transversely on the posterolateral neck at C2-C4. Depth ≈ 1.5-3 cm.

## Obliquus Capitis Inferior obliquus-capitis-inferior

Cervical dystonia — ipsilateral head rotation (torticollis)

-  **Patient Position** pos-cervical-posterior-flexed.jpg  Position the patient prone or seated with the neck slightly flexed and the forehead supported to open the suboccipital space.
-  **Probe + Needle Site** obliquus-capitis-inferior-probe.jpg Patient prone/seated, neck slightly flexed. Identify the C2 spinous process, then angle obliquely toward the C1 transverse... Needle: Under ultrasound, identify the muscle spanning C2 spinous to C1 transverse process and insert in-plane toward the belly.
-  **Ultrasound Image** obliquus-capitis-inferior-us.jpg Identify the bifid C2 spinous process in the midline as the bony landmark. Depth ≈ 2-4 cm (deep).

## Semispinalis Capitis semispinalis-capitis

Bilateral head and neck extension (retrocollis)

-  **Patient Position** pos-cervical-posterior-flexed.jpg  Position the patient seated with the neck slightly flexed to open the posterior cervical space, or prone with the forehead resting on hands.
-  **Probe + Needle Site** semispinalis-capitis-probe.jpg Patient seated, neck flexed. Place probe transversely on posterior neck at C2-C5 level, 1-2 cm lateral to midline. Photo from... Needle: This is a DEEP muscle — the needle must traverse the trapezius and splenius capitis to reach it. USG guidance is essential to...
-  **Ultrasound Image** semispinalis-capitis-us.jpg Identify the skin and subcutaneous fat as the most superficial hyperechoic layer.

## Longissimus Capitis longissimus-capitis

Head and neck extension (retrocollis), ipsilateral lateral flexion and rotation

-  **Patient Position** pos-cervical-posterior-flexed.jpg  Position the patient seated with the neck slightly flexed to open the posterior cervical space, or prone with the forehead resting on hands.
-  **Probe + Needle Site** longissimus-capitis-probe.jpg Patient seated, neck slightly flexed. Place probe transversely on posterolateral neck at C4-C7 level, 2-4 cm lateral to midline.... Needle: This is a DEEP muscle in the erector spinae group — the needle must traverse the trapezius and splenius capitis to reach it. USG...
-  **Ultrasound Image** longissimus-capitis-us.jpg Identify the skin and subcutaneous fat as the most superficial hyperechoic layer.



## Frontalis frontalis not typically US-guided

*Brow elevation dystonia, forehead spasm, involuntary brow raising; NOTE: this is primarily a cosmetic target (horizontal forehead lines) and is NOT a typical spasticity target*

### ☐ Patient Position pos-seated-face-forward.jpg shared

Position the patient seated upright. Ask them to raise their eyebrows forcefully — the frontalis will contract, creating horizontal forehead lines.

### ☐ Probe + Needle Site frontalis-probe.jpg

Landmark-guided injection — no ultrasound used. Patient seated upright, raising eyebrows to activate frontalis. Four to five... Needle: Standard cosmetic approach: inject at 4-5 sites across the mid-forehead, each approximately 1.5-2 cm above the brow and spaced... No US — red dot only; blue bar optional.

### ☐ Ultrasound Image frontalis-us.jpg

Optional — not typically ultrasound-guided.

## Corrugator Supercilii corrugator-supercilii not typically US-guided

*Blepharospasm (glabellar component), brow furrowing, glabellar dystonia*

### ☐ Patient Position pos-seated-face-forward.jpg shared

Position the patient seated upright. Ask them to frown forcefully — the corrugator creates the vertical (11 lines) furrows between the brows.

### ☐ Probe + Needle Site corrugator-supercilii-probe.jpg

Landmark-guided injection — no ultrasound used. Patient seated upright, frowning to activate the corrugator. Injection at the... Needle: Insert a 30G needle at a shallow angle (20-30 degrees from the skin surface) directed laterally along the corrugator muscle... No US — red dot only; blue bar optional.

### ☐ Ultrasound Image corrugator-supercilii-us.jpg

Optional — not typically ultrasound-guided.

## Procerus procerus not typically US-guided

*Blepharospasm (glabellar component), horizontal nasal root furrows, glabellar dystonia*

### ☐ Patient Position pos-seated-face-forward.jpg shared

Position the patient seated upright. Ask them to 'scrunch' their nose or pull their brows down toward the nose — the procerus creates horizontal...

### ☐ Probe + Needle Site procerus-probe.jpg

Landmark-guided injection — no ultrasound used. Patient seated upright, actively scrunching the nose to identify the procerus.... Needle: Insert a 30G needle perpendicular to the skin or at a shallow angle into the midline nasal root/glabella junction, advancing only... No US — red dot only; blue bar optional.

### ☐ Ultrasound Image procerus-us.jpg

Optional — not typically ultrasound-guided.

## Orbicularis Oculi orbicularis-oculi not typically US-guided

*Blepharospasm (involuntary forceful eye closure), hemifacial spasm*

### ☐ Patient Position pos-seated-face-forward.jpg shared

Position the patient seated upright in a well-lit room. The patient should be relaxed with eyes gently closed or in their resting blepharospasm...

### ☐ Probe + Needle Site orbicularis-oculi-probe.jpg

Landmark-guided injection — no ultrasound used. Patient seated upright in well-lit room. Six injection sites distributed around... Needle: Insert the needle subcutaneously (bevel up) at each site, directed AWAY from the globe. The muscle is extremely thin (1-2 mm) and... No US — red dot only; blue bar optional.

### ☐ Ultrasound Image orbicularis-oculi-us.jpg

Optional — not typically ultrasound-guided.

## Temporalis temporalis

*Jaw clenching, bruxism, oromandibular dystonia (jaw closing type), temporal headache*

### ☐ Patient Position pos-seated-jaw-relaxed.jpg shared

Position the patient seated upright or supine with the jaw relaxed.

### ☐ Probe + Needle Site temporalis-probe.jpg

Patient seated upright with jaw relaxed. Place probe transversely in the temporal fossa above the zygomatic arch. Photo from... Needle: Insert the needle perpendicular to the skull surface into the muscle belly at each site. Advance only 0.5-1 cm — the temporalis...

### ☐ Ultrasound Image temporalis-us.jpg

Place the probe transversely on the temporal fossa above the zygomatic arch. Apply minimal pressure — the temporalis is thin and... Depth ≈ 1.0-1.5 cm.

## Masseter masseter

*Jaw clenching, bruxism, oromandibular dystonia (jaw closing type)*

### ☐ Patient Position pos-seated-jaw-relaxed.jpg shared

Position the patient seated upright or supine with the jaw relaxed and slightly open.

### ☐ Probe + Needle Site masseter-probe.jpg

Patient seated upright with jaw relaxed. Place probe transversely on lateral face over the mandibular ramus midway between... Needle: Ask the patient to clench their teeth forcefully — the masseter will bulge prominently as a thick rectangular mass over the...

### ☐ Ultrasound Image masseter-us.jpg

Place the probe transversely on the lateral face over the mandibular ramus, midway between the zygomatic arch and the angle of... Depth ≈ 2.0-3.0 cm.

## Lateral Pterygoid lateral-ptyergoid not typically US-guided

*Jaw opening dystonia, jaw protrusion, oromandibular dystonia (jaw opening type), jaw deviation to contralateral side*

### ☐ Patient Position pos-seated-jaw-relaxed.jpg shared

Position the patient seated upright or supine with the jaw slightly open (to translate the condyle forward and open the coronoid notch).

### ☐ Probe + Needle Site lateral-ptyergoid-probe.jpg

EMG-guided injection — no ultrasound probe placement. Patient seated with jaw slightly open. Needle inserted through coronoid... Needle: Insert a 25-27G, 1.5-2 inch needle through the coronoid notch, directing it medially and slightly anteriorly toward the... No US — red dot only; blue bar optional.

### ☐ Ultrasound Image lateral-ptyergoid-us.jpg

Optional — not typically ultrasound-guided.

## Medial Pterygoid medial-ptyergoid

*Jaw clenching, oromandibular dystonia (jaw closing type), synergist with masseter for jaw elevation*

### ☐ Patient Position pos-seated-jaw-relaxed.jpg shared

Position the patient seated upright or supine with the jaw slightly open.

### ☐ Probe + Needle Site medial-ptyergoid-probe.jpg

Patient seated with jaw slightly open. For US guidance: place probe in submandibular region below mandibular angle, angled... Needle: SUBMANDIBULAR (EXTRAORAL) APPROACH (preferred for US guidance): Palpate the medial surface of the mandibular angle from the...

### ☐ Ultrasound Image medial-ptyergoid-us.jpg

Place the probe in the submandibular region below and medial to the angle of the mandible, angled superiorly. Depth ≈ 2.5-4.0 cm.

**Mentalis** mentalis not typically US-guided

*Chin dimpling dystonia, involuntary chin puckering, mental crease deepening, lower lip protrusion*

 **Patient Position** pos-seated-face-forward.jpg shared

Position the patient seated upright. Ask them to 'pout' or push the lower lip forward — the mentalis will contract, producing the characteristic...

 **Probe + Needle Site** mentalis-probe.jpg

Landmark-guided injection — no ultrasound used. Patient seated upright, pouting to activate the mentalis and display chin... Needle: For unilateral dystonia or dimpling, inject the affected side only. For bilateral chin dimpling, inject both sides. No US — red dot only; blue bar optional.

 **Ultrasound Image** mentalis-us.jpg

Optional — not typically ultrasound-guided.

**Platysma** platysma not typically US-guided

*Anterocollis (forward head flexion), neck banding, platysmal dystonia, involuntary neck tightening*

 **Patient Position** pos-seated-face-forward.jpg shared

Position the patient seated upright or supine. Ask them to grimace or depress the mandible with lips retracted to make the platysmal bands prominent.

 **Probe + Needle Site** platysma-probe.jpg

Landmark-guided injection — platysmal bands are identified visually by asking the patient to grimace. Patient seated upright,... Needle: Insert a 30G needle subcutaneously into each platysmal band, directing it along the length of the band. Advance only 2-3 mm — the... No US — red dot only; blue bar optional.

 **Ultrasound Image** platysma-us.jpg

Optional — not typically ultrasound-guided.